



Horizontal Pod Autoscaler (HPA)



- Horizontal Pod Autoscaler (HPA) is a native Kubernetes object that automatically scales the number of pods in a replication controller, deployment, replica set, or stateful set
- HPA is based on observed
 - CPU utilization
 - custom metrics support like memory
- The Metrics Server in Kubernetes cluster is used by HPA to track resource demands of pods
- The number of pods is automatically increased whenever an application requires more resources.
The number of pods is decreased when demand gets lower
- HPA helps to ensure that applications have sufficient resources to handle varying workloads
- HPA reduces the need for manual intervention in scaling applications
- HPA helps in achieving better resource utilization and cost-effectiveness